

A Doctoral Thesis Title

by

Your Full Name

A thesis submitted for the degree of

Doctor of Philosophy

Data Lab for Social Good Research Group

Cardiff University, United Kingdom

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Abstract

Insert a summary of no more than 300 words here. This should briefly state the research problem, aims, methods, main findings, and contribution of the thesis.

Acknowledgements

Integer id risus vel lorem laoreet commodo lobortis quis purus. Cras cursus leo vel dui laoreet pulvinar. Nunc tincidunt metus et ante fermentum lacinia. Proin quam magna, tristique ut viverra at, dapibus eget elit. Quisque eu leo id nisi semper laoreet at ac nulla. Fusce volutpat, metus sed dictum mattis, nisl elit dapibus velit, non porttitor urna urna vel diam. Praesent tortor nulla, rutrum ac magna a, tempor sagittis enim. Praesent pharetra ipsum libero, eu malesuada libero blandit ut. Sed sed venenatis ligula, nec convallis turpis. Nulla iaculis felis eros, eget pharetra lorem cursus quis. Nunc iaculis lobortis magna at malesuada. Nullam elementum elit at urna congue aliquam.

Declaration

This thesis is the result of my own independent work, except where otherwise stated, and the views expressed are my own. Other sources are acknowledged by explicit references. The thesis has not been edited by a third party beyond what is permitted by Cardiff University's Use of Third-Party Editors by Research Degree Students Procedure.

This thesis is constructed as a series of papers. Chapters 2, 3, and 4 should therefore be read as separate entities. The appendices relating to each of these papers are included as separate appendices at the end of the thesis.

The word count for this thesis is

Signed:

Date:

Statement of Authorship

The paper(s) listed below have been included in the thesis.

- The paper(s) report on original research I conducted during my research degree candidature.
- Published and accepted papers have not been modified, other than their formatting.
- Current publication status is stated (published work / accepted for publication / submitted for publication / unpublished work in manuscript style and of publishable quality).
- My contribution is described, and a percentage contribution is estimated.
- Where a paper has been published or is accepted for publication, the publisher has given permission for the paper to be included, and for the thesis to become available subject to any embargo in accordance with the Policy and Procedure on Bars on Access to Research Degree Theses.

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(by typing your name, you are providing your signature)			

Montana, E., Itala, H. and Costata, V. (2023) [paper title and details]

Publication status: **published work**

Publisher's permission: **granted**

Presented as Chapter 3

EM contribution: 80%

EM developed the methodology with the assistance of IH and VC, and carried out the analysis. EM led the writing of the paper, with input from IH and VC, and advice from reviewers.

Montana, E. and Itala, H. (2024) [paper title and details]

Publication status: **accepted for publication.**

Publisher's permission: **granted**.

Presented as Chapter 4.

EM contribution: 90%.

EM designed the study with the assistance of IH, and carried out the analysis. EM drafted the paper; IH reviewed and commented on the paper prior to its submission to the journal.

Montana, E., Hammonis, N. and Costata, V. (submitted) [paper title]

Publication status: **unpublished and unsubmitted work in manuscript style**.

Publisher's permission: **N/A**.

Presented as Chapter 5.

EM contribution: 60%.

NH and VC designed the model; EM and NH designed the study with the assistance of VC. The model simulations were completed by EM and NH. EM carried out the analysis and led the writing of the paper, with input from both co-authors.

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List of Abbreviations

LAH List **A**bbreviations **H**ere

WSF **W**hat (it) **S**tands **F**or

List of Symbols

a	distance	m
P	power	W (J s ⁻¹)
ω	angular frequency	rad

“For Elsa ”

Chapter 1: Getting Started with the Template

1.1 Overview

This template provides a structured workflow for writing a Cardiff University thesis in Quarto and rendering it to PDF using LuaLaTeX. It is designed to give a clear starting point for thesis writing while keeping the writing process simple, modular, and reproducible. The template separates content from formatting so that chapters, references, appendices, and front matter can be managed in an organised way.

The thesis is built as a Quarto book project. This means that the main project configuration is defined in `_quarto.yml`, while each chapter is written in a separate `.qmd` file. Front matter elements such as acknowledgements, declaration, abbreviations, symbols, and dedication can also be stored separately. The overall formatting of the PDF output is controlled through the custom class file together with the LaTeX partial files.

This chapter introduces the structure of the template and explains where the main files are located. Chapter 2 then explains how to work with citations, figures, tables, equations, references, and appendices in practice.

1.2 What This Template Includes

The template is organised around a small number of core folders and files.

1.2.1 Main files

The main project configuration file is `_quarto.yml`. This defines the thesis metadata, output settings, and the order of chapters and appendices. The `index.qmd` file acts as the opening chapter of the thesis. The bibliography file is `references.bib`, which stores all references cited in the thesis. The custom PDF formatting is controlled through `MastersDoctoralThesis.cls`, together with the LaTeX partial files in the `partials/` folder.

1.2.2 Main folders

The `Chapters/` folder contains the main thesis chapters. Each chapter should be written in its own `.qmd` file. The `Appendices/` folder contains appendix files. The `Frontmatter/` folder contains optional front matter files such as acknowledgements,

declaration, abbreviations, symbols, and dedication. Figures may either be generated directly from code chunks or stored separately in a folder such as `Figures/`.

1.3 How the Thesis Is Structured

The thesis follows a standard structure consisting of front matter, main chapters, references, and appendices. The front matter is generated before the main body begins and may include the title page, abstract, declaration, acknowledgements, table of contents, list of figures, list of tables, and optional lists such as abbreviations and symbols. The main body consists of the numbered chapters. The references are placed at the end of the thesis, followed by any appendices.

This structure is controlled through the YAML in `_quarto.yml`, where the order of chapters and appendices is listed explicitly.

1.4 Editing the Metadata

Most of the thesis information is set in `_quarto.yml`. This is where the title, author name, submission date, degree name, university, bibliography file, output settings, and chapter order are defined. This is also where optional front matter files can be included or excluded.

As a result, one of the first steps in using the template is to replace the default metadata with your own details and check that the listed chapter files match the structure you intend to use.

1.5 Writing Chapters

Each thesis chapter should be written in a separate `.qmd` file using normal Quarto markdown headings. For example, a chapter should begin with a top-level heading using `#`, followed by sections using `##` and subsections using `###`. Section numbers should not be typed manually because Quarto generates them automatically when section numbering is enabled.

This structure keeps each chapter manageable and makes the full thesis easier to maintain. It also helps ensure that chapter numbering, cross-references, and table of contents entries update correctly when the thesis is re-rendered.

1.6 Front Matter Files

Front matter files are optional and are usually stored in the `Frontmatter/` folder as separate `.tex` files. These can include items such as acknowledgements, declaration, dedication, abbreviations, and symbols. They are then linked through `_quarto.yml` and inserted automatically when the thesis is rendered.

If the thesis is written as a traditional monograph, no paper-specific authorship statement is needed unless required by the School or University. If the thesis is written in a paper-based format, or includes jointly authored published, accepted, or submitted papers, then an authorship or contribution statement should be included where appropriate. If the thesis is not paper-based, this section should be omitted.

1.7 Why This Template Uses Quarto

Quarto provides a convenient writing environment that combines plain-text authoring with high-quality PDF output. It also supports references, figures, tables, equations, and cross-references in a consistent way. For thesis writing, this is especially useful because it allows the thesis to remain modular while still rendering as a single document.

The template therefore aims to provide a practical balance between flexibility and structure. The main goal is not to hide the underlying mechanics completely, but to reduce the amount of formatting work that has to be done while writing.

As with other reproducible writing workflows, the separation of source files, bibliography, and formatting makes revision easier over time (Hawthorn, Weber, and Scholten 2001; Wieman and Hollberg 1991).

1.8 Before You Begin Writing

Before drafting chapters, it is worth checking the overall structure of the thesis and ensuring that the project metadata, bibliography file, and chapter order are correct. It is also useful to render the template once at an early stage to confirm that the local Quarto and LaTeX setup is working properly.

Once that is done, the thesis can be developed chapter by chapter within the project structure.

1.9 Concluding Remarks

This chapter has introduced the overall purpose and structure of the template. The next chapter explains the practical workflow in more detail, including how to add chapters, insert figures and tables, cite references, and work with appendices.

Chapter 2: Using the Template in Practice

2.1 Writing Workflow

The template is designed around a simple workflow. Each chapter is written in its own file, references are managed through the bibliography database, and the final thesis is rendered from the project root directory. This makes the writing process more modular than working in a single long document and reduces the risk of formatting changes affecting the whole thesis unexpectedly.

In practice, the most common workflow is to edit one chapter file at a time, render the full thesis periodically, and use the compiled PDF to check headings, numbering, references, figures, and overall layout.

2.2 Adding Chapters

New chapters should be added as separate `.qmd` files inside the `Chapters/` folder. Once a chapter file has been created, it should be listed in the `chapters:` section of `_quarto.yml`. The order in which the files are listed there determines the order in which they appear in the thesis.

Each chapter should begin with a top-level heading. For example, `# Introduction` will produce a numbered chapter heading in the final PDF. Lower heading levels should then be used to organise sections and subsections within the chapter.

2.3 Cross-Referencing

Quarto provides built-in support for cross-references. This allows figures, tables, and equations to be labelled and cited consistently throughout the thesis.

Figures should be labelled using the `fig-` prefix, tables using the `tbl-` prefix, and equations can be given their own labels. Once labelled, they can be referred to in the text using the relevant reference key. This makes it easier to revise the document later because numbering is updated automatically.

For example, Figure Figure 2.1, Table Table 2.1, and Equation Equation 2.1 below are all cross-referenced from within this chapter.

2.4 Citations and Bibliography

All references cited in the thesis should be stored in `references.bib`. In the body text, citations are inserted using citation keys. In this template, the bibliography keys are assumed to follow a format such as `Reference1`, `Reference2`, and so on.

For example, `@Reference1` will generate an in-text citation, while `[@Reference1; @Reference2]` will generate a parenthetical citation containing more than one source. The bibliography is generated automatically and placed at the end of the thesis.

This keeps citation management separate from the main text and makes it easier to revise or replace sources later. This is particularly useful in long-form academic writing projects (Hawthorn, Weber, and Scholten 2001).

2.5 Example Figure

The figure below illustrates a simple render cycle.

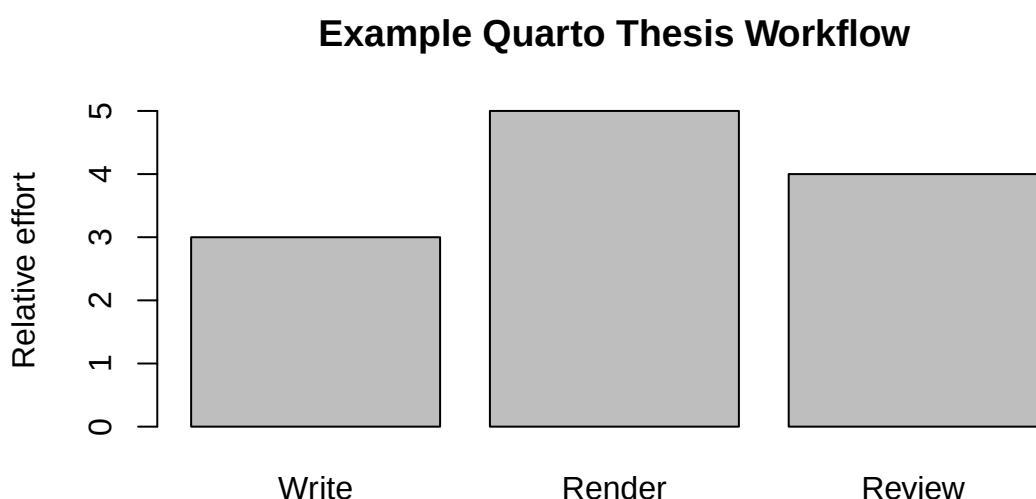


Figure 2.1: Example rendering cycle for working with the template.

Figure 2.1 gives a simple illustration of a write-render-review workflow.

2.6 Example Table

The following table shows examples of common reference syntax in the template.

Table 2.1: Example citation and cross-reference patterns.

Item	Example
In-text citation	Hawthorn, Weber, and Scholten (2001)
Parenthetical citation	(Hawthorn, Weber, and Scholten 2001; Wieman and Hollberg 1991)
Figure reference	Figure 2.1
Table reference	Table 2.1
Equation reference	Equation 2.1

Table 2.1 provides a small summary of common reference patterns.

2.7 Example Equation

Equations can be inserted and labelled directly in the text.

$$R = W + C + E \quad (2.1)$$

Equation Equation 2.1 is only a placeholder example, but it shows the syntax used for numbering and cross-referencing equations.

2.8 Figures and Tables from External Files

Figures can also be inserted from saved image files rather than generated through code. In that case, image files may be stored in a folder such as `Figures/` and inserted using standard Quarto image syntax. Tables can similarly be generated from imported data or prepared directly in code chunks.

This flexibility makes it possible to combine static content and reproducible analysis outputs within the same thesis.

2.9 Front Matter

The template supports several front matter elements, including the abstract, declaration, acknowledgements, dedication, abbreviations, and symbols. It also supports the table of contents, list of figures, and list of tables. These elements are stored separately from the main body so that the structure remains clear. Some of them are optional and can be included or removed depending on the needs of the thesis.

If the thesis is paper-based, or includes jointly authored published, accepted, or

submitted papers, an authorship or contribution statement can also be included where appropriate. If the thesis is not paper-based, this should be omitted.

The order of the front matter is handled by the template partial files and related thesis settings rather than by the chapter files themselves.

2.10 References and Appendices

The references are intended to appear at the end of the thesis. A dedicated `references.qmd` file can be used to place the bibliography at the correct point in the document. Appendices then follow after the references.

If appendices are used, they should be placed in the `Appendices/` folder and listed in the `appendices:` section of `_quarto.yml`. Quarto then handles the appendix structure in the rendered thesis.

2.11 Rendering the Thesis

The thesis should be rendered from the project root. In most cases, the simplest way is to run `quarto preview` during writing and `quarto render` when producing a final PDF. Rendering the whole project rather than individual chapter files ensures that chapter numbering, front matter, references, and appendices are all handled correctly.

This is especially important for a thesis template because section numbering and book structure depend on Quarto recognising the project as a complete book.

2.12 Practical Advice

A useful habit is to render early and often. Small structural issues such as broken cross-references, missing bibliography keys, or misplaced appendices are easier to correct when noticed early. It is also good practice to keep chapters modular, store all references in the bibliography file, and use Quarto's built-in labelling system consistently.

In a long thesis project, these small habits make the workflow much more stable (Wieman and Hollberg 1991).

References

- Hawthorn, C. J., K. P. Weber, and R. E. Scholten. 2001. "Littrow Configuration Tunable External Cavity Diode Laser with Fixed Direction Output Beam." *Review of Scientific Instruments* 72 (12): 4477–79. <http://link.aip.org/link/?RSI/72/4477/1>.
- Wieman, Carl E., and Leo Hollberg. 1991. "Using Diode Lasers for Atomic Physics." *Review of Scientific Instruments* 62 (1): 1–20. <http://link.aip.org/link/?RSI/62/1/1>.

Appendix A: Frequently Asked Questions

A.1 How do I change the colors of links?

Pass in `urlcolor:` in yamf. Or set these in the include-in-header file.

If you want to completely hide the links, you can use:

```
{\hypersetup{allcolors=.}}, or even better:
```

```
{\hypersetup{hidelinks}}.
```

If you want to have obvious links in the PDF but not the printed text, use:

```
{\hypersetup{colorlinks=false}}.
```